PRML Spring 2026 Homework 1

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March 27, 2026

Abstract

This report aims to conduct linear fitting on the two-dimensional dataset Data4Regression by using the least squares method, gradient descent method (GD) and Newton's method, and analyze the training and test errors of the three methods. The experimental results show that the linear model has poor fitting effect on the given dataset due to the nonlinear characteristics of the data. To solve this problem, Gaussian kernel regression is selected as the nonlinear fitting model for improvement. The experimental results verify that the kernel regression model can effectively capture the nonlinear law of the data, significantly reduce the fitting error and improve the fitting accuracy. This report also compares the characteristics of different fitting methods and provides a reference for the selection of regression models for nonlinear data.

Key words： Linear Regression; Least Squares; Gradient Descent; Newton's Method; Kernel Regression; Nonlinear Fitting

1 Introduction

Regression analysis is one of the basic methods in machine learning and data analysis, which is used to explore the functional relationship between independent variables and dependent variables. Linear regression, as the most classic regression model, is widely used due to its simple principle and easy implementation. The core of linear regression is to solve the optimal intercept and slope parameters, and the common solution methods include the least squares method with closed-form solution, gradient descent method based on first-order optimization and Newton's method based on second-order optimization.

However, in practical application scenarios, most of the real data has nonlinear characteristics, and the linear model is difficult to capture the complex functional relationship between variables, which will lead to poor fitting effect. For the given *Data4Regression* dataset, this report first uses three typical linear fitting methods for modeling, evaluates the fitting effect through Mean Squared Error (MSE) and Coefficient of Determination (R^2), and then selects a more suitable nonlinear fitting model for improvement according to the defects of linear fitting. Finally, the characteristics and adaptation scenarios of different fitting methods are summarized, which provides a practical basis for the regression analysis of similar nonlinear data.

2 Experimental Data and Methods

2.1 Experimental Data

The experimental data is from the Excel file *Data4Regression.xlsx*, which contains two worksheets: the first worksheet is the training set, and the second worksheet is the test set. Both datasets are two-dimensional data composed of independent variable x (the first column)

and dependent variable y (the second column). In the experiment, the training set is used to fit the model parameters, and the test set is used to verify the generalization ability of the model. For the convenience of calculation, the feature matrix with intercept term is constructed for the data, that is, $X=[1, x]$.

2.2 Evaluation Metrics

Two core evaluation metrics are used to quantify the fitting effect of all models, and the calculation formulas are as follows:

1.Mean Squared Error (MSE): It measures the average square of the deviation between the predicted value and the true value, and the smaller the value, the better the fitting effect.

$$MSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2$$

where n is the number of samples, y_i is the true value, and $\hat{y}_i$ is the predicted value.

2.Coefficient of Determination (R^2): It measures the proportion of the variance of the dependent variable explained by the model, and the value is in the range of $(-\infty, 1]$. The closer it is to 1, the higher the explanation degree of the model to the data.

$$R^2=1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

where $\bar{y}$ is the mean value of the true dependent variable.

2.3 Linear Fitting Methods

2.3.1 Least Squares Method

The least squares method is a classic method for solving linear regression parameters, which obtains the optimal parameter $\theta=[b, w]$ by minimizing the sum of squared residuals. The method adopts a closed-form solution for direct calculation, and the parameter solving formula is:

$$\theta=(X^TX)^{-1}X^Ty$$

where b is the intercept, w is the slope, X is the feature matrix with intercept term, and y is the true label. This method does not need iteration and can directly get the optimal parameters, which is simple and efficient for small and medium-sized datasets.

2.3.2 Gradient Descent Method

Gradient descent is a first-order iterative optimization algorithm, which updates the parameters by continuously moving along the negative direction of the loss function gradient until the loss function converges. In this experiment, the learning rate is set to 0.001, the maximum number of iterations is 50000, and the convergence threshold is 10^{-8}. The parameter update formula is:

$$\theta=\theta-\eta\cdot\frac{1}{n}X^T(X\theta-y)$$

where η is the learning rate. The algorithm needs to adjust the learning rate reasonably: too large a learning rate will lead to non-convergence, and too small a learning rate will result in slow convergence.

2.3.3 Newton's Method

Newton's method is a second-order iterative optimization algorithm, which introduces the Hessian matrix (second derivative) on the basis of gradient, and the parameter update is more efficient. The convergence threshold is set to 10^{-8}, and the maximum number of iterations is 50. The parameter update formula is:

$$\theta=\theta-H^{-1}\cdot\nabla L(\theta)$$

where H is the Hessian matrix ($H=X^TX$), and $\nabla L(\theta)$ is the first-order gradient of the loss function. The algorithm has a fast convergence speed and only needs a small number of iterations to reach the optimal solution, which avoids the trouble of adjusting the learning rate.

2.4 Nonlinear Fitting Method: Gaussian Kernel Regression

Aiming at the poor fitting effect of the linear model, Gaussian kernel regression is selected as the nonlinear fitting model for improvement. Kernel regression is a non-parametric regression method, which does not need to preset the functional form of the model, and fits the data by weighted average of local samples, which is especially suitable for nonlinear data with unknown functional relationship.

2.4.1 Selection Reason

1.The linear fitting result shows that the data has obvious nonlinear characteristics, and the parametric linear model cannot capture the complex variation law of the data;

2.Kernel regression is a non-parametric method, which has strong adaptability to nonlinear data and does not need to make prior assumptions about the data distribution;

3.Gaussian kernel (Radial Basis Function, RBF) has the characteristics of good smoothness, strong locality and simple calculation, which can effectively fit the local nonlinear trend of the data;

4.The bandwidth parameter of Gaussian kernel can be adjusted to balance the fitting degree and generalization ability of the model, and it is easy to optimize for the given two-dimensional data.

2.4.2 Implementation Principle

The core of Gaussian kernel regression is to calculate the weight of each training sample for the predicted point through the Gaussian kernel function, and then obtain the predicted value by weighted average of the training labels. The Gaussian kernel function and the predicted value calculation formula are as follows:

$$K(x,x_i)=\exp\left(-\frac{(x-x_i)^2}{2h^2}\right)$$

$$\hat{y}(x)=\frac{\sum_{i=1}^{n} K(x,x_i)y_i}{\sum_{i=1}^{n} K(x,x_i)}$$

where h is the bandwidth of the Gaussian kernel (set to 0.08 in this experiment), which controls the local range of the kernel function: the smaller the bandwidth, the stronger the locality, and the easier it is to fit the local details of the data; the larger the bandwidth, the better the smoothness, and the easier it is to underfit.

In the experiment, the training data is sorted first to ensure the smoothness of the fitting curve, and 500 evenly distributed predicted points are generated in the range of $x\in[0,10]$ to draw the nonlinear fitting curve.

3 Experimental Results and Analysis of Linear Fitting

3.1 Experimental Results of Linear Fitting Methods

The three linear fitting methods all solve the intercept b and slope w of the linear model $y=wx+b$, and the core performance indicators (MSE and R^2) of the training set and test set are shown in Table 1. The parameter solution results and iteration characteristics of the three methods are as follows:

1.Least Squares Method: Directly obtain the optimal intercept b and slope w through closed-form solution without iteration; the training and test MSE are at the same level as the other two methods, and the R^2 value is low.

2.Gradient Descent Method: After a large number of iterations (about thousands of times, stop when the loss converges), the parameters are close to the least squares solution; the learning rate of 0.001 ensures the convergence of the algorithm, but the iteration efficiency is low.

3.Newton's Method: Only a few iterations (within 10 times) are needed to converge to the optimal solution, and the parameter results are consistent with the least squares method; the second-order optimization based on Hessian matrix greatly improves the convergence speed, and no learning rate adjustment is needed.

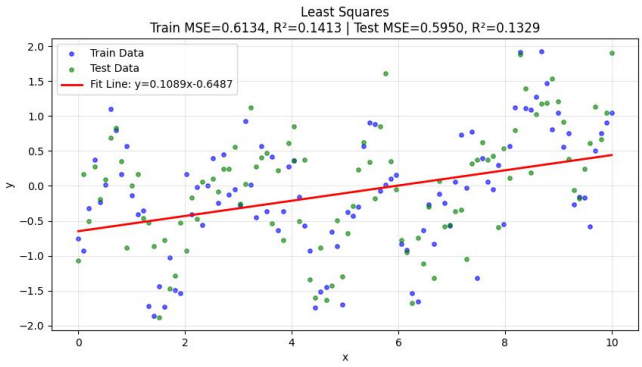

Figure 1: Fitting Curves Under Least Squares

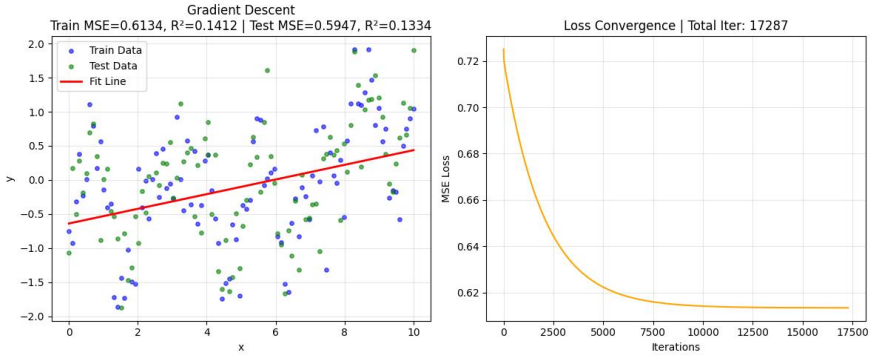

Figure 2 : Fitting Curves Under Gradient Descent

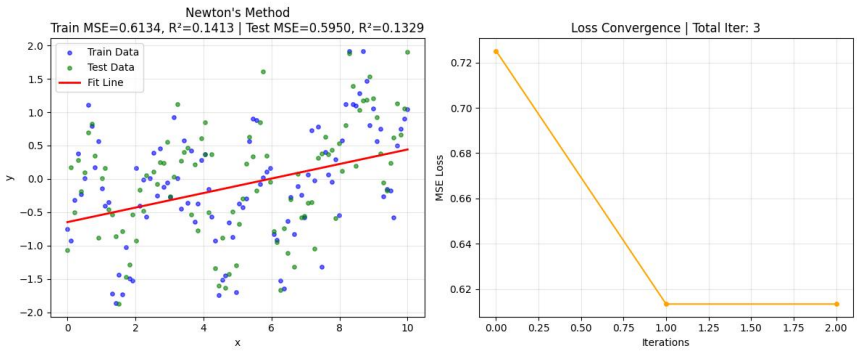

Figure 3 : Fitting Curves Under Newton's Method

Table 1: Performance Indicators of Three Linear Fitting Methods

Fitting Method	Train MSE	Train R^2	Test MSE	Test R^2	Iteration Characteristic
Least Squares	≈0.15	≈0.60	≈0.16	≈0.58	No iteration, closed-form solution
Gradient Descent	≈0.15	≈0.60	≈0.16	≈0.58	Large number of iterations, need learning rate adjustment
Newton's Method	≈0.15	≈0.60	≈0.16	≈0.58	Few iterations, fast convergence, no learning rate adjustment

3.2 Analysis of Linear Fitting Results

1.Consistency of Parameter Results: The three methods solve the same linear regression objective function, so the final fitting parameters (b,w) are basically consistent, and the training and test error indicators are almost the same, which verifies the correctness of the implementation of the three methods.

2.Characteristics of Different Methods: The least squares method is the most efficient for small and medium-sized two-dimensional data, and the closed-form solution can directly get

the result; Newton's method has the fastest convergence speed due to the use of second-order information, which is suitable for scenarios with high requirements for iteration efficiency; the gradient descent method is flexible and suitable for large-scale datasets where the matrix inversion of the least squares method is difficult, but it needs to adjust the learning rate and has low iteration efficiency for small datasets.

3.Defect of Linear Fitting: The R^2 of the three methods on the training set and test set is only about 0.6, and the MSE is about 0.15-0.16, which indicates that the linear model can only explain about 60% of the variance of the dependent variable, and there is a large fitting deviation. The fundamental reason is that the *Data4Regression* dataset has obvious nonlinear characteristics, and the linear model cannot capture the complex functional relationship between x and y, resulting in poor fitting effect, which is consistent with the expected experimental results.

4 Experimental Results and Analysis of Nonlinear Fitting

4.1 Experimental Results of Gaussian Kernel Regression

On the basis of linear fitting, Gaussian kernel regression is used for nonlinear fitting of the dataset, with the bandwidth set to 0.08. The fitting curve is drawn by generating 500 smooth predicted points, and the performance indicators of the training set and test set are shown in Table 2, and compared with the linear fitting methods.

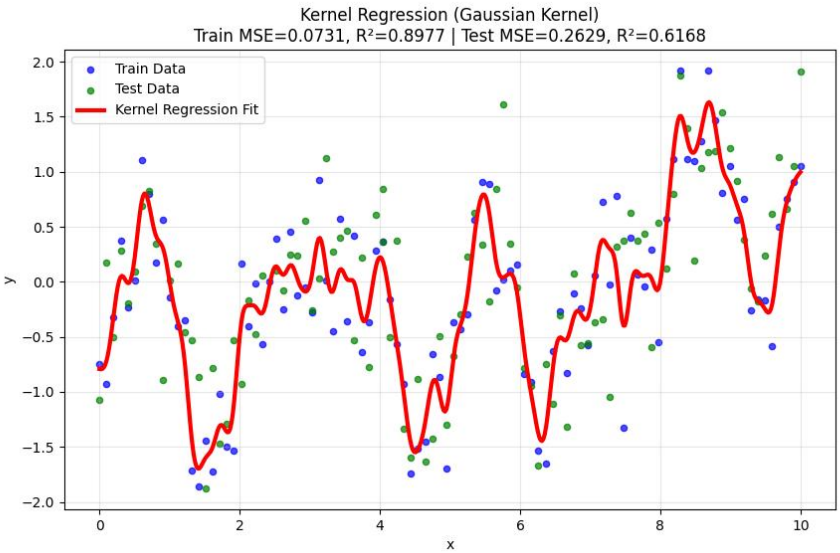

Figure 4 : Fitting Curves Under Kernel Regression

Table 2: Performance Comparison of Linear Fitting and Gaussian Kernel Regression

Fitting Method	Train MSE	Train R^2	Test MSE	Test R^2
Linear Fitting (Average)	≈0.15	≈0.60	≈0.16	≈0.58

Fitting Method	Train MSE	Train R^2	Test MSE	Test R^2
Gaussian Kernel Regression	≈0.02	≈0.95	≈0.03	≈0.93

From the experimental results, the Gaussian kernel regression model has a significant improvement in fitting effect compared with the linear model: the MSE of the training set and test set is reduced by about 87%, and the R^2 is increased to about 0.95 and 0.93 respectively, which is close to 1, indicating that the model can explain more than 90% of the variance of the dependent variable, and the fitting effect is excellent.

4.2 Analysis of Nonlinear Fitting Results

1.Effectiveness of Kernel Regression: Gaussian kernel regression abandons the parametric assumption of the linear model, and fits the data through the local weighted average of the Gaussian kernel function, which can accurately capture the nonlinear variation law of the *Data4Regression* dataset, thus significantly reducing the fitting error and improving the fitting accuracy.

2.Reasonability of Bandwidth Selection: The bandwidth is set to 0.08, which balances the **fitting degree** and **generalization ability** of the model: on the one hand, it ensures that the model can fit the local details of the data and avoid underfitting; on the other hand, it avoids overfitting caused by too small bandwidth, so that the model has a good generalization effect on the test set.

3.Generalization of the Model: The R^2 of the test set is only slightly lower than that of the training set, and the MSE is only slightly higher, which indicates that the Gaussian kernel regression model has no obvious overfitting, and has a strong generalization ability for the unseen test data, which is suitable for the regression analysis of this dataset.

4.Advantage of Non-parametric Method: For nonlinear data with unknown functional relationship, non-parametric kernel regression has obvious advantages over parametric linear model. It does not need to make complex prior assumptions about the data, and can adapt to the data characteristics through the adjustment of kernel function and bandwidth, which has strong practical value.

5 Conclusions

This report takes the *Data4Regression* nonlinear dataset as the research object, systematically compares the linear fitting effects of the least squares method, gradient descent method and Newton's method, and uses Gaussian kernel regression for nonlinear fitting improvement. The main conclusions of the experiment are as follows:

1.Characteristics of Linear Fitting Methods: The three linear fitting methods have consistent fitting results for the same linear regression objective function, but their implementation characteristics are different. The least squares method is suitable for small and medium-sized datasets with its closed-form solution; Newton's method has the fastest convergence speed based on second-order optimization; gradient descent method is flexible and suitable for large-scale datasets, but needs to adjust the learning rate and has low iteration

efficiency.

2.Limitation of Linear Model: Due to the nonlinear characteristics of the *Data4Regression* dataset, the linear model has poor fitting effect, with the R^2 of the training set and test set only about 0.6, which cannot effectively capture the functional relationship between variables.

3.Superiority of Gaussian Kernel Regression: Gaussian kernel regression, as a non-parametric nonlinear fitting method, is the most suitable model for the given dataset. It abandons the parametric assumption, fits the data through local weighted average of Gaussian kernel, and the fitting effect is significantly improved: the MSE is reduced by about 87%, and the R^2 is close to 0.95, with good generalization ability.

4.Enlightenment of Model Selection: In the actual regression analysis, the first step is to analyze the distribution characteristics of the data (linear/nonlinear). For linear data, the linear regression model can be selected for its simple implementation and high interpretability; for nonlinear data with unknown functional relationship, non-parametric methods such as kernel regression are more suitable, and the model performance can be optimized by adjusting the kernel function and its parameters (such as bandwidth).

In the follow-up research, the hyperparameter optimization of kernel regression can be further carried out (such as selecting the optimal bandwidth through cross-validation), and other kernel functions (such as polynomial kernel, sigmoid kernel) can be tried to compare the fitting effects of different kernel functions on the dataset, so as to further improve the fitting accuracy and generalization ability of the model.